

Wi-Fi Series

Data Sheet

USE_8851

Wi-Fi / BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	17/07/2026	MB	AD	Initial release

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Features :

General

- Supports Wi-Fi and Bluetooth functionalities.
- Wi-Fi 6 dual-band 1T1R module.
- Bluetooth 5.3 compliant.
- PCIe interface for WLAN and USB interface for Bluetooth.
- Enhanced BT/WIFI Coexistence Control to improve transmission quality in different profiles
- Integrated 32K oscillator for power management
- Dimension: 12 × 16 × 1.95mm, typical.
- Operating temperature: -10°C to 70°C.
- Storage temperature: -55°C to 125°C.
- OS supported : Android /Linux/ Windows

MCU Platform

- Chipset : RTL8851BE-CG.

WLAN Features

- IEEE 802.11a/b/g/n/ac/ax compatible WLAN.
- Supports 20MHz and 40MHz channel bandwidth at 2.4GHz.
- Supports 20MHz, 40MHz and 80MHz channel bandwidth at 5GHz.
- Supports IEEE 802.11ac 1×1 Wave-2 with RX MU-MIMO.
- Maximum PHY data rate up to 150Mbps with IEEE 802.11n, 433Mbps with IEEE 802.11ac and 600Mbps with IEEE 802.11ax.
- Supports single-antenna or dual-antenna designs with diversity.

Host Interface

- Complies with PCI Express Base Specification Revision 1.1 for WLAN.
- Complies with USB 2.0 Full-Speed specification for Bluetooth.

Bluetooth Features

- Support Bluetooth 5 system (BT 5.3 Logo compliant).
- Compatible with Bluetooth v2.1 + EDR.
- Dual Mode support: simultaneous LE and BR/EDR.
- Integrated internal Class 1, Class 2 and Class 3 power amplifiers.

General Description:

The USE_8851 is a highly integrated Wi-Fi and Bluetooth module based on the Realtek RTL8851BE-CG chipset. It combines a WLAN MAC, 1T1R-capable WLAN baseband, RF circuitry and Bluetooth functionality in a compact module.

The module supports IEEE 802.11a/b/g/n/ac/ax wireless LAN standards and Bluetooth 5.3. WLAN communication is provided through a PCI Express interface, while Bluetooth communication is provided through a USB interface.

The USE_8851 supports one transmit and one receive path. It supports MU-MIMO OFDM, Short Guard Interval, spatial spreading, STBC, Transmit Beamforming and LDPC. The module can be used with a single antenna or two antennas with diversity.

The module supports DSSS, CCK and OFDM baseband processing for compatibility with legacy IEEE 802.11a/b/g standards. It supports BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM and 1024-QAM modulation.

For IEEE 802.11ax operation, the module supports OFDMA reception with RU26, RU52, RU106, RU242, RU484 and RU996 resource unit sizes. The module supports a maximum PHY data rate of up to 600Mbps.

The module includes enhanced signal detection, adaptive frequency-domain equalization and interference suppression to reduce the effects of multipath interference, Bluetooth interference, cordless phone interference and microwave oven interference.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

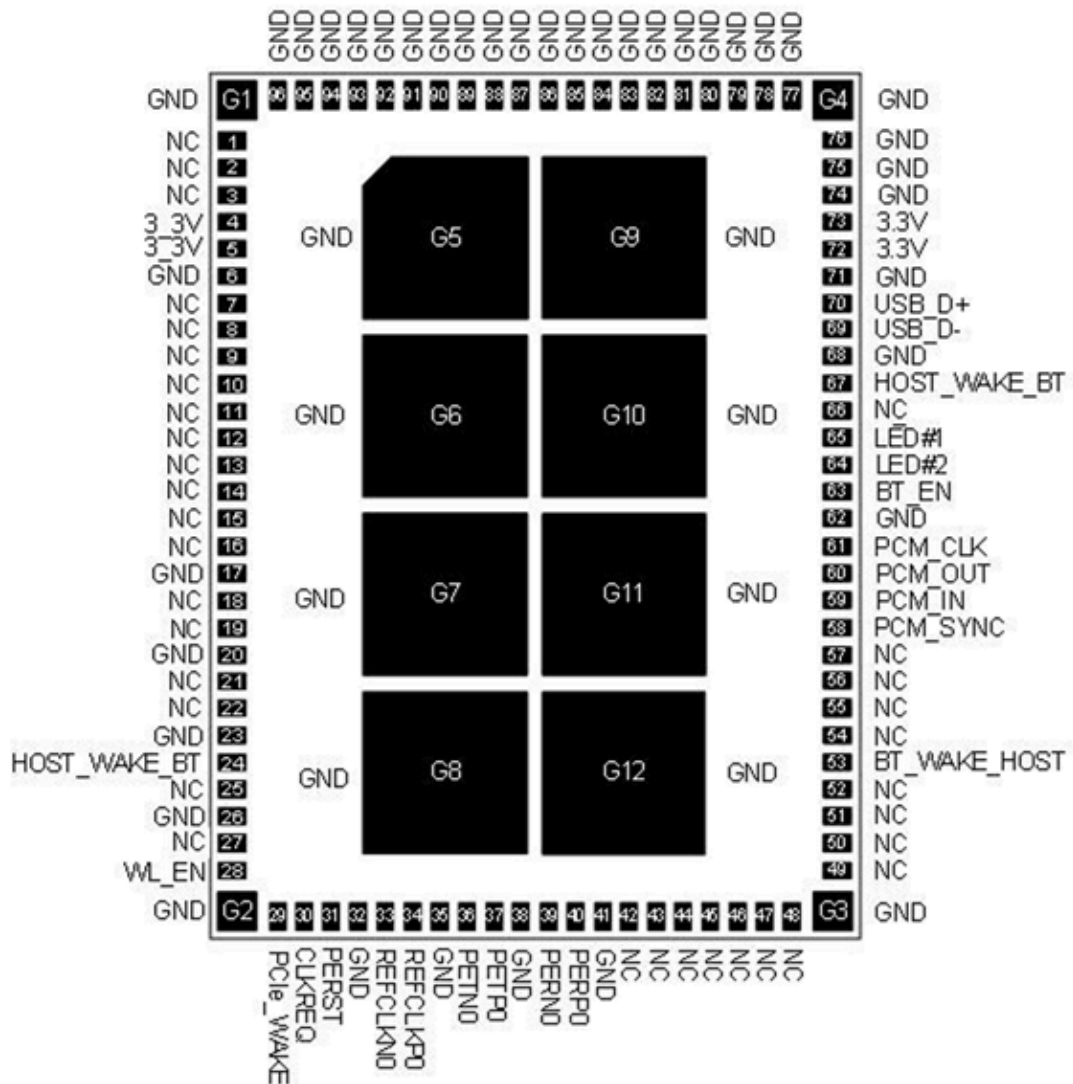


Fig1: Module Pinouts

Pin Descriptions:

Pin No.	Pin Name	Type	Description
1	NC	—	No connect
2	NC	—	No connect
3	NC	—	No connect

4	3_3V	P	Main power voltage source input(3.3V)
5	3_3V	P	Main power voltage source input(3.3V)
6	GND	—	Ground connection
7	NC	—	No connect
8	NC	—	No connect
9	NC	—	No connect
10	NC	—	No connect
11	NC	—	No connect
12	NC	—	No connect
13	NC	—	No connect
14	NC	—	No connect
15	NC	—	No connect
16	NC	—	No connect
17	GND	—	Ground connection
18	NC	—	No connect
19	NC	—	No connect
20	GND	—	Ground connection
21	NC	—	No connect
22	NC	—	No connect
23	GND	—	Ground connection
24	HOST_WAKE_BT	I	Host wake-up Bluetooth
25	NC	—	No connect
26	GND	—	Ground connection

27	NC	—	No connect
28	WL_EN	—	WLAN enable pin. High: enable; Low: disable
29	PCle_WAKE	OD	PCle wake-up host, open-drain, active low
30	CLKREQ	OD	PCle reference-clock request signal
31	PERST	PD	PCle reset module
32	GND	—	Ground connection
33	REFCLKN0	I	PCle differential clock negative
34	REFCLKP0	I	PCle differential clock positive
35	GND	—	Ground connection
36	PETN0	O	PCle differential data output negative
37	PETP0	O	PCle differential data output positive
38	GND	—	Ground connection
39	PERN0	I	PCle differential data input negative
40	PERP0	I	PCle differential data input positive
41	GND	—	Ground connection
42	NC	—	No connect
43	NC	—	No connect
44	NC	—	No connect
45	NC	—	No connect
46	NC	—	No connect
47	NC	—	No connect
48	NC	—	No connect
49	NC	—	No connect

50	NC	—	No connect
51	NC	—	No connect
52	NC	—	No connect
53	BT_WAKE_HOST	O	Bluetooth wake-up host
54	NC	—	No connect
55	NC	—	No connect
56	NC	—	No connect
57	NC	—	No connect
58	PCM_SYNC	I/O	PCM synchronization signal
59	PCM_IN	I	PCM data input
60	PCM_OUT	O	PCM data output
61	PCM_CLK	I/O	PCM sync clock
62	GND	—	Ground connection
63	BT_EN	I	Bluetooth enable pin. Pulling this pin low shuts down the Bluetooth function and disables the USB interface
64	LED#2	O	BT link LED, active low.
65	LED#1	O	WLAN link LED, active low
66	NC	—	No connect
67	HOST_WAKE_BT	I	Host wake-up Bluetooth, active high
68	GND	—	Ground connection
69	USB_D-	I/O	USB differential data line negative for Bluetooth
70	USB_D+	I/O	USB differential data line positive for Bluetooth

71	GND	—	Ground connection
72	3.3V	P	Main power voltage source input 3.3V
73	3.3V	P	Main power voltage source input 3.3V
74–76	GND	—	Ground connections
G1–G12	GND	—	Ground connections

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	-10	25	70	°C
VDD33	3.0	3.3	3.6	V

RF Characteristics

1. 2.4G RF Specification

Test Conditions: VBAT = 3.3V, VDDIO = 3.3V, Temperature = 25°C

Feature	Description	
WLAN Standard	IEEE 802.11 b/g/n/ac/ax Wi-Fi compliant	
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz: Ch1 ~ Ch14	
Test Items	Typical Value	EVM
	802.11b /11Mbps : 19dBm ± 1.5 dB	EVM £ -9dB
	802.11g /54Mbps : 18dBm ± 1.5 dB	EVM £ -25dB
	802.11n /MCS7 : 17dBm ± 1.5 dB	EVM £ -28dB
	802.11ac VHT20/MCS8: 16dBm ± 1.5dB	EVM £ -30dB

Output Power ¹	802.11ac VHT40/MCS9: 15dBm $\pm$ 1.5 dB	EVM $\leq$ -32dB
	802.11ax HE20/MCS11: 13dBm $\pm$ 1.5 dB	EVM $\leq$ -35dB
	802.11ax HE40/MCS11: 13dBm $\pm$ 1.5 dB	EVM $\leq$ -35dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	$\pm$ 20ppm	
SISO Receive Sensitivity (11b,20MHz) @8% PER	- 1Mbps @ -94 dBm	$\leq$ -83 dBm
	- 11Mbps @ -85 dBm	$\leq$ -76 dBm
SISO Receive Sensitivity (11g,20MHz) @10% PER	- 6Mbps @ -90 dBm	$\leq$ -85 dBm
	- 54Mbps @ -71 dBm	$\leq$ -68 dBm
SISO Receive Sensitivity (11n,20MHz) @10% PER	- MCS=0 @ -90 dBm	$\leq$ -85 dBm
	- MCS=7 @ -69 dBm	$\leq$ -67 dBm
SISO Receive Sensitivity (11n,40MHz) @10% PER	- MCS=0 @ -87 dBm	$\leq$ -82 dBm
	- MCS=7 @ -66 dBm	$\leq$ -64 dBm
SISO Receive Sensitivity (11ac,20MHz) @10% PER	- MCS=0 @ -90 dBm	$\leq$ -82 dBm
	- MCS=8 @ -64 dBm	$\leq$ -60 dBm
SISO Receive Sensitivity(11ac ,40MHz) @10% PER	- MCS=0 @ -87 dBm	$\leq$ -79 dBm
	- MCS=9 @ -59 dBm	$\leq$ -55 dBm
SISO Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=0 @ -90 dBm	$\leq$ -74 dBm
	- MCS=11 @ -60 dBm	$\leq$ -52 dBm
SISO Receive Sensitivity (11ax ,40MHz) @10% PER	- MCS=0 @ -87 dBm	$\leq$ -71 dBm
	- MCS=11 @ -57 dBm	$\leq$ -49 dBm
802.11b : -10 dBm		

Maximum Input Level	802.11g/n : -20 dBm
Antenna Reference	Small antennas with 0~2 dBi peak gain

2. 5GHz RF Specification

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

Feature	Description	
WLAN Standard	IEEE 802.11a/n/ac/ax, Wi-Fi compliant	
Frequency Range	5.150 GHz ~ 5.850 GHz (5 GHz Band)	
Number of Channels	5.0GHz: Please see the table ¹	
Test Items	Typical Value	EVM
Output Power ²	802.11a /54Mbps: 18 dBm $\pm$ 1.5 dB	EVM $\leq$ -25dB
	802.11n /MCS7: 17 dBm $\pm$ 1.5 dB	EVM $\leq$ -28dB
	802.11ac VHT20/MCS8: 16 dBm $\pm$ 1.5 dB	EVM $\leq$ -30dB
	802.11ac VHT40/MCS9: 15 dBm $\pm$ 1.5 dB	EVM $\leq$ -32dB
	802.11ac VHT80/MCS9: 15 dBm $\pm$ 1.5 dB	EVM $\leq$ -32dB
	802.11ax HE20/MCS11: 13 dBm $\pm$ 1.5 dB	EVM $\leq$ -35dB
	802.11ax HE40/MCS11: 13 dBm $\pm$ 1.5 dB	EVM $\leq$ -35dB
	802.11ax HE80/MCS11: 13 dBm $\pm$ 1.5 dB	EVM $\leq$ -35dB
Test Items	Test Value	Standard Value
SISO Receive Sensitivity	- 6Mbps @ -90 dBm	$\leq$ -85 dBm
(11a,20MHz) @10% PER	- 54Mbps @ -71 dBm	$\leq$ -68 dBm
SISO Receive Sensitivity (11n,20MHz) @10% PER	- MCS=0 @ -90 dBm	$\leq$ -85 dBm
	- MCS=7 @ -69 dBm	$\leq$ -67 dBm

SISO Receive Sensitivity (11n,40MHz) @10% PER	- MCS=0 @ -87 dBm	≤ -82 dBm
	- MCS=7 @ -66 dBm	≤ -64 dBm
SISO Receive Sensitivity (11ac,20MHz)@10% PER	- MCS=0, NSS1 @ 90 dBm	≤ -82 dBm
	- MCS=8, NSS1 @ -64 dBm	≤ -60 dBm
SISO Receive Sensitivity (11ac,40MHz) @10% PER	- MCS=0, NSS1 @ -87 dBm	≤ -79 dBm
	- MCS=9, NSS1 @ -59 dBm	≤ -55 dBm
SISO Receive Sensitivity (11ac,80MHz) @10% PER	- MCS=0, NSS1 @ -84 dBm	≤ -79 dBm
	- MCS=9, NSS1 @ -56 dBm	≤ -54 dBm
SISO Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=0 @ -90 dBm	≤ -74 dBm
	- MCS=11 @ -60 dBm	≤ -52 dBm
SISO Receive Sensitivity (11ax,40MHz) @10% PER	- MCS=0 @ -87 dBm	≤ -71 dBm
	- MCS=11 @ -57 dBm	≤ -49 dBm
SISO Receive Sensitivity (11ax,80MHz) @10% PER	- MCS=0 @ -84 dBm	≤ -68 dBm
	- MCS=11 @ -54 dBm	≤ -46 dBm
Maximum Input Level	802.11a/n: -30 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

- 2.4G,5G output power control by firmware power by rate table

3. 5GHz(20MHz) Channel table

Band range	Operating Channel Numbers	Channel center frequencies(MHz)
5150MHz~5250MHz	36	5180
	40	5200

	44	5220
	48	5240
5250MHz~5350MHz	52	5260
	56	5280
	60	5300
	64	5320
5470MHz~5725MHz	100	5500
	104	5520
	108	5540
	112	5560
	116	5580
	120	5600
	124	5620
	128	5640
	132	5660
	136	5680
	140	5700
5725MHz~5850MHz	149	5745
	153	5765
	157	5785
	161	5805
	165	5825

4. Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	BDR,EDR(1Mbps & 2Mbps & 3Mbps),LE(1Mbps),2LE(2Mbps)
Host Interface	USB

Frequency Band	2400 MHz ~ 2483.5MHz
Number of Channels	79 channels for classic,40 channels for BLE
Modulation	GFSK, $\pi/4$ -DQPSK,8DPSK
RF Specification	
Output Power , tolerance ± 3 dB	
	CL1(dBm)
BDR Output Power	5
EDR Output Power	5
BLE Output Power	5
Sensitivity, tolerance : /	
Sensitivity @ BER=0.1% for GFSK (1Mbps)	-70
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)	-70
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)	-70
Sensitivity @ BLE=30.8% for LE (1Mbps)	-70
Sensitivity @ BLE=30.8% for 2LE (2Mbps)	-70
Maximum Input Level	GFSK (1Mbps):-20dBm
	$\pi/4$ -DQPSK (2Mbps) :-20dBm
	8DPSK (3Mbps) :-20dBm

Block Diagram :

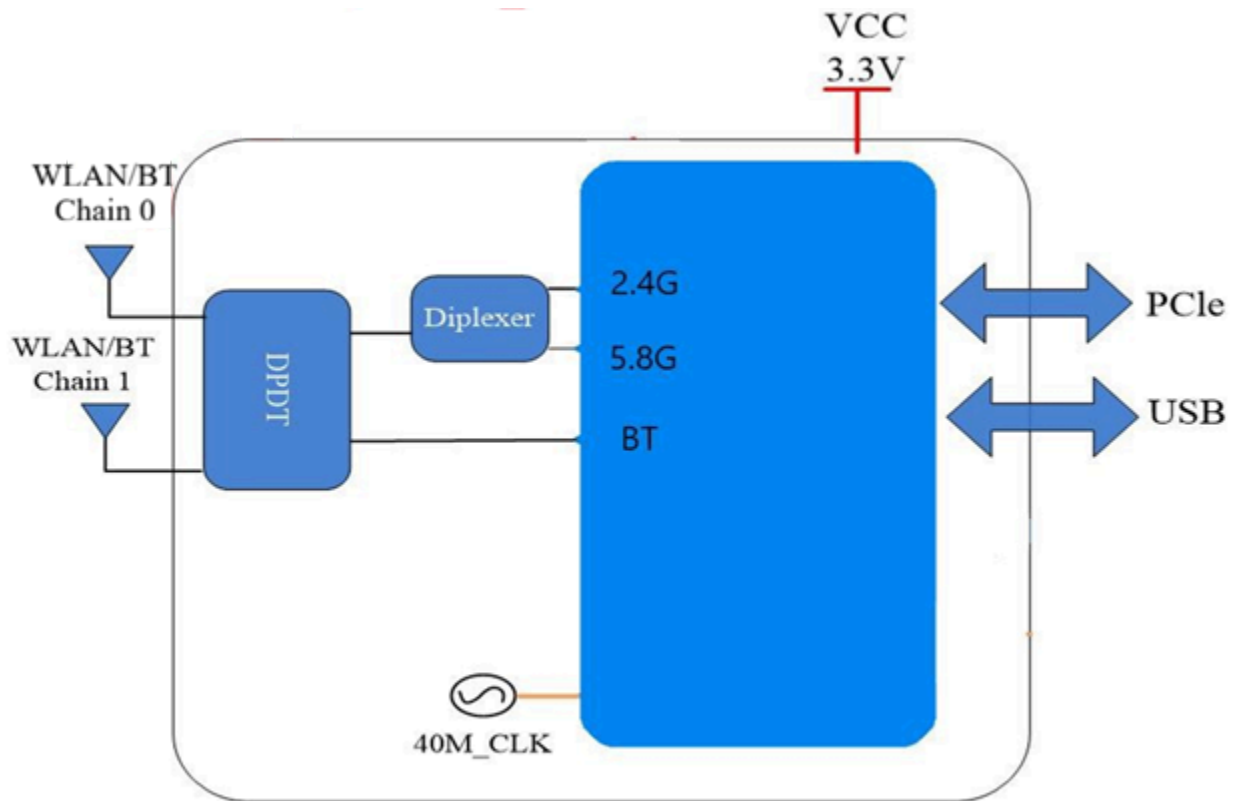


Fig 2 : Block Diagram

Reference Design :

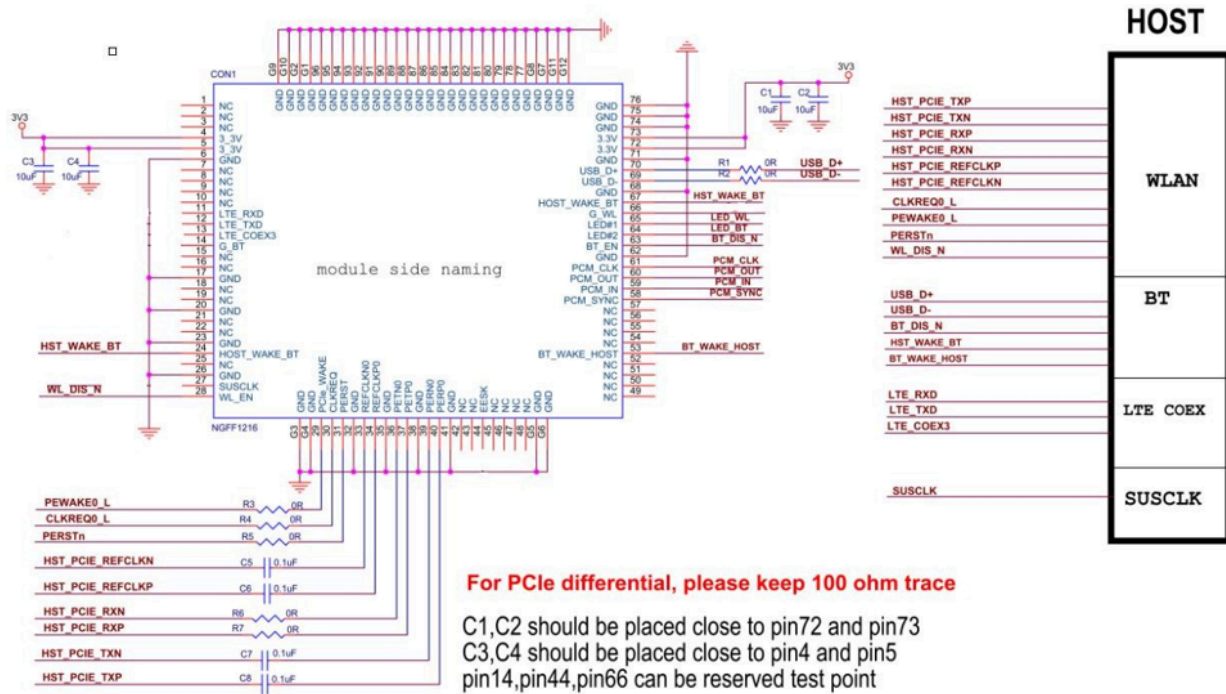


Fig 3 : Reference Design

Note:

- Both antenna ports support 2.4GHz Wi-Fi, 5GHz Wi-Fi and Bluetooth operation.
- The USE_8851 antenna port is controlled by the driver when the diversity function is enabled
- PCIe differential traces shall maintain 100Ω differential impedance.
- USB differential traces shall maintain 90Ω differential impedance.

Recommended Footprint Dimensions:

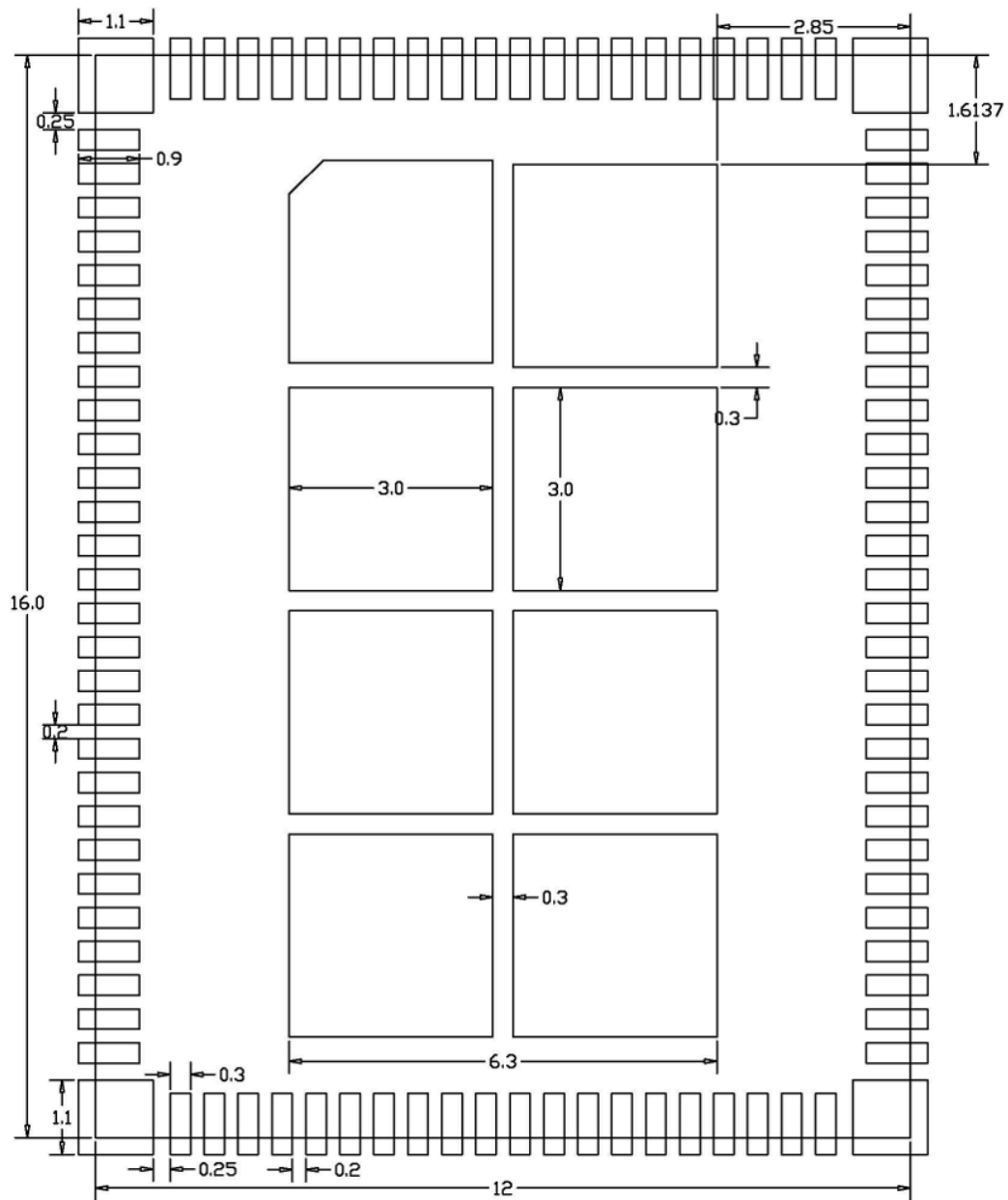


Fig 4 : Recommended Footprint Dimensions

Module Dimension Drawing:

- Module Dimension: L x W : 12 x 16mm

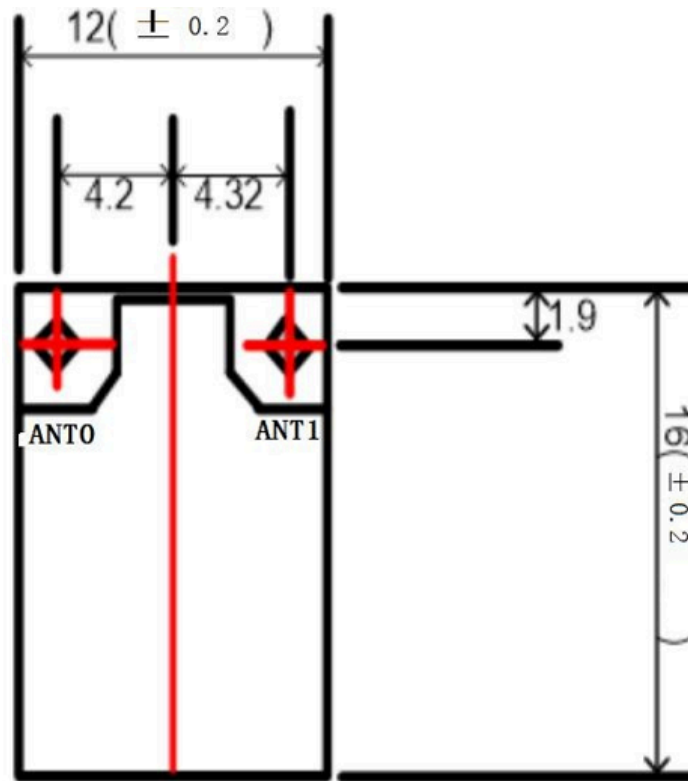


Fig 5(a) Module Dimension L × W

Product Dimension:



Fig 5(b) :Product Dimension

PCB thickness : 0.6mm

Product Height (Including shield) : 1.95mm

Weight : 0.807g

Recommended Reflow Profile :

Referred to IPC/JEDEC standard.

Peak Temperature: $260 \pm 5^{\circ}\text{C}$

Time within 5°C of peak temperature : $\geq 10\text{s}$

Number of Times: 2 times

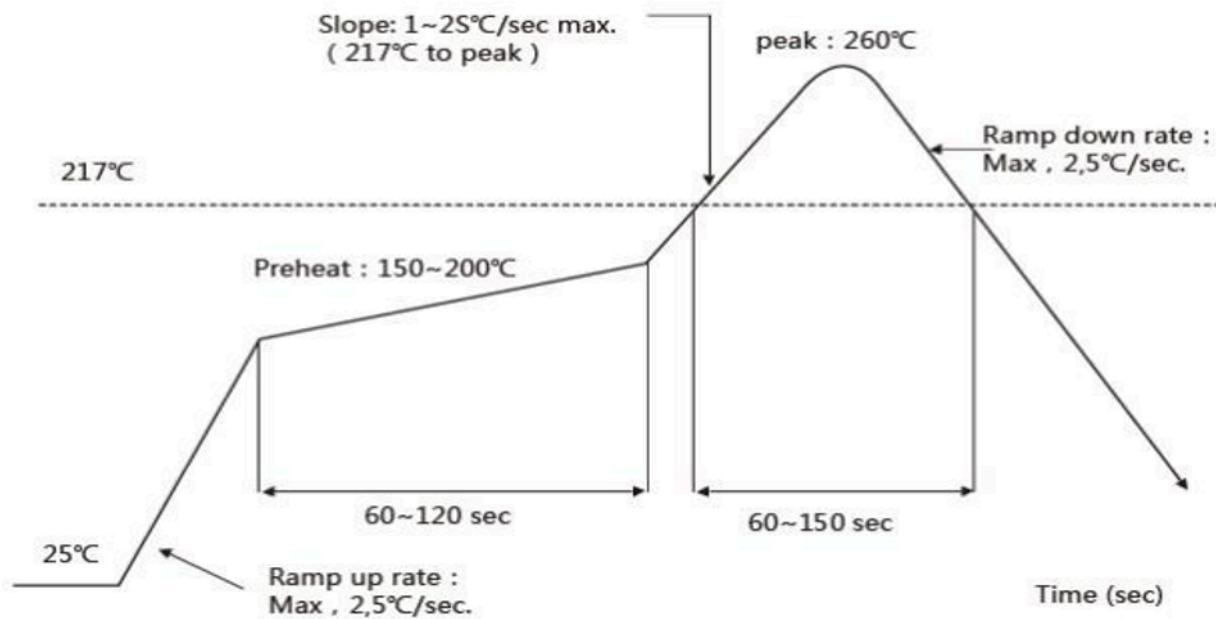
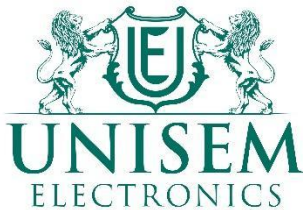


Fig 6 : Recommended Reflow Profile

For ordering USE_8851 please contact us on info@unisemindia.com.



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